Software Requirements Specification

for IntelliLib:

Library Management System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Library Management System	17.04.2026	-	1.0

1. Introduction

1.1 Purpose

The primary objective of the Library Management System (LMS) is to establish an efficient, structured, and intuitive web-based platform for managing comprehensive library operations within an academic institution. This system is designed to streamline the interaction between Librarians and Members (students/staff) by automating critical functions such as book issuance, return processing, catalog searching, and detailed record administration.

The core goal of the system is to substantially enhance the accuracy, speed, and accessibility of library services while simultaneously mitigating manual effort, reducing reliance on physical paperwork, and minimizing errors inherent in traditional library management methodologies.

The LMS provides users with the capability to browse the collection, monitor the status of borrowed materials, and maintain library records effectively through a centralized digital infrastructure.

1.2 Document Conventions

This Software Requirements Specification (SRS) document adheres to the IEEE standard format for software requirement documentation.

- All requirements are presented from a functional perspective.
- Standard software engineering terminology is consistently employed throughout the document.
- Bullet points and structured formatting are utilized to enhance clarity and readability.
- Consistent naming conventions, such as *Librarian* and *Member*, are maintained for system components.

1.3 Intended Audience and Reading Suggestions

1.3.1 Intended Audience

- End Users: Students and staff members who will utilize the system for searching, requesting, and borrowing library resources.
- Librarians: Administrative personnel responsible for the management of books, user accounts,

and library transactions.
- Institutions: Universities or organizations requiring a centralized and efficient system for managing library operations.
- Technical Stakeholders: Developers, system analysts, and quality assurance personnel who require a comprehensive understanding of system functionality and imposed constraints.

1.3.2 Reading Guidance

- General users are advised to concentrate on the Introduction and Product Scope sections to comprehend the system's intended usage and associated benefits.
- Librarians should meticulously review the features and functionalities pertinent to administrative duties.
- Developers and evaluators are required to read the entirety of this document to gain a full understanding of the system's design, specified requirements, and constraints.

1.4 Product Scope

The system ensures better data organization, quick access, transparency, and improved user experience compared to traditional systems.

The Library Management System provides a centralized digital platform to manage all library activities effectively.

The system enables the following functionalities:

1.4.1 For Librarian:

- Issue books to members.
- Accept returned books.
- Calculate and collect late fines.
- Add new books and maintain book records.
- Edit information of existing books.
- View categories of books.
- View lists of books under each category.
- Generate reports of available books.
- Generate reports of issued books.
- Access and manage all member accounts.

1.4.2 For Members:

- View book categories and available books.
- Search for specific books.
- Maintain a personal account.
- View issued books.
- View accumulated late fines.

- Request new books.
- View history of previously issued books.

1.4.3 Additions:

- Book Availability Control: Prevents over-issuance by initiating availability checks by the Librarian.
- User Issue Limits: Enforces configurable limits (set by the Librarian) on the total number of books a Member may borrow.
- Late Fine Management: Calculates applicable fines and may impose borrowing restrictions; involves the Member (fine payment responsibility) and is managed by the Librarian.
- Book Renewal (Extension): Processes a Member's extension request, which is approved or rejected by the Librarian.
- Multiple Copy Tracking: Manages inventory and monitors availability of multiple copies; Librarian updates status, while the Member can only view availability.
- Audit Log Records: Maintains a record of all activities for accountability, accessible exclusively by the Librarian; Members are prohibited from access.

1.5 References

None

2. Overall Description

2.1 Product Perspective

The Library Management System (LMS) is a web-based, stand-alone application designed to manage and automate library operations within an institution such as a university, college, or organization. The system provides a centralized digital platform where Librarians and Members (students/staff) can interact efficiently.

The system allows users to browse books, search for items, issue and return books, and maintain records in a structured and organized manner. It replaces traditional manual record-keeping systems and enhances them by providing faster access, better data management, and improved accuracy.

The LMS does not depend on any existing automated system but can integrate with institutional databases if required. The Librarian acts as the system controller, managing all activities including book records, user accounts, and reports.

2.2 Product Functions

The main functions of the Library Management System include:

Use Case	Description
Sign Up	Allows new members to create an account in the system
Login	Enables users (members/librarian) to securely log in
View Book Categories	Displays different categories of books available in the library
View Book List	Shows all books available under each category
Search Book	Allows users to search for specific books
Issue Book	Librarian issues books to members
Return Book	Librarian accepts returned books
Add Book	Librarian adds new books to the database
Edit Book Details	Librarian updates existing book information
View Issued Books	Members can view books currently issued to them
Book Request	Members can request new books

Use Case	Description
View History	Members can view previously issued books
Generate Reports	Librarian can view reports of available and issued books
Notifications	Sends alerts for availability, claims, and system updates
Manage Accounts	Librarian can access and manage member accounts
Late Fine Calculation	System automatically calculates fines for overdue books
Fine Payment	Members can view and pay pending fine
Fine Management	Librarian can update fine rules and track payments
Logout	Allows users to securely log out of the system
Book Availability Tracking	Tracks and displays real-time availability of books to prevent issuing unavailable copies.
Issue Limit Control	Restricts the maximum number of books a member can borrow at a given time.
Fine Restriction Control	Prevents users with unpaid fines above a defined limit from issuing new books.
Book Renewal (Extension)	Allows members to request an extension of the due date for issued books, subject to approval.
Book Copy Management	Manages multiple copies of the same book and updates their availability dynamically.
Role-Based Access Control	Ensures that system features are accessible only to users with appropriate roles and permissions.
Activity Logging (Audit Logs)	Records all important system activities for monitoring, tracking, and accountability.

2.3 User Classes and Characteristics

The different user classes involved in the Library Management System are:

User Class	Description
Members	Can search books, issue/return, view categories, request books, and track issued books, view history, and pay late fines
Librarian	Manages books, issues/returns, users, reports, and monitors fine rules and

	payment and system monitoring through logs

2.4 Operating Environment

2.4.1 Software Requirements:

- Compatibility is mandated with contemporary web browsers, specifically Google Chrome, Mozilla Firefox, Microsoft Edge, and Internet Explorer.
- Operation shall be facilitated on web-based platforms accessible through a standard web browser.
- Supported operating systems shall include Windows (designated as the primary platform), in addition to other operating systems that furnish requisite browser support.

2.4.2 Hardware Requirements:

- Hard Disk Drive: A minimum capacity of 256 GB is required.
- Random Access Memory (RAM): A minimum of 4 GB is recommended for optimal performance.
- Display Unit: A 15.-6 inch color monitor is required.
- Input Peripherals: Keyboard and Mouse are necessary.
- Output Peripherals: Display unit and a Printer are required.
- A persistent and reliable internet connection is a prerequisite for system functionality.

2.5 Design and Implementation Constraints

2.5.1 Platform, Connectivity, and Performance

- Implementation: The system shall be implemented as a web-based application.
- Connectivity: Continuous, stable internet connectivity is mandatory for system operation.
- Performance: System performance is directly dependent on the capabilities of the hosting server and the network transmission speed.

2.5.2 Financial Management and Configuration

- Fine Calculation: The system must accurately calculate monetary fines based on established due dates.
- Fine Configurability: Authorized librarians shall have the ability to modify fine calculation rules (e.g., the accrued amount per day).
- Borrowing Restriction: The system must prevent members from borrowing books if their pending fines exceed a defined financial threshold.
- Payment Records: All payment transaction records must be securely stored.

2.5.3 Security and Access Control

- Access Control: Access to the system is strictly limited to authorized personnel, including registered members and librarians.
- Audit Logs: Audit logs must be secure and tamper-resistant.

2.5.4 Data Management

- Data Storage: All system data shall be securely stored using a centralized database architecture.
- Data Integrity: The system is mandated to ensure data integrity and consistency across all procedures and transactions.
- Book Inventory: The system must maintain an accurate count of available copies for all books.

2.5.5 Usability and Business Rules

- User Interface: The user interface shall be simple, intuitive, and easy to operate.
- Issue Limits: The system must enforce established limits on the number of books a user can issue.
- Renewal Validation: All book renewal requests must be validated before approval.

2.6 User Documentation

The system provides fundamental user guidance through several methods:

- In-System Help: Users can access help sections and instructions directly on various web pages.
- Workflow Guidance: Specific instructions are available for core operations, including logging in, searching for books, and performing issuing and returning transactions.
- Management Assistance: Guidance is offered for tasks related to managing user accounts and viewing reports.

2.7 Assumptions and Dependencies

This section lists the assumptions and external dependencies that may affect the requirements and behavior of the Library Management system.

2.7.1 Assumptions

- User Environment: Users are operating with a stable internet connection and a compatible web browser on a functional computer.
- Data Integrity: All information supplied and maintained by users during system interactions is assumed to be accurate and valid.
- Librarian Responsibilities: Library staff will precisely manage book records and execute all transaction processes accurately.

- System Availability: The system is expected to maintain continuous (24/7) and uninterrupted operation.
- System Quality: The system is anticipated to be highly intuitive (user-friendly) and free from technical defects.
- Fine Policy Adherence: Users are expected to comply with established issue limits and renewal policies.
- Fine Calculation: Fines are determined by predefined, configurable rules, such as a set daily penalty rate.
- Fine Clearance Requirement: Members must settle any outstanding fines before being authorized to issue additional books.
- Time Accuracy: The system's date and time are presumed to be accurate, which is crucial for correct fine and due date calculation.

2.7.2 Dependencies

- Database: A reliable, operational database is mandatory for the secure storage of all book and user data, and must support real-time updates for availability and logging.
- Network Connectivity: Continuous internet access is essential for system functionality.
- Performance: System performance is directly dependent on the server's efficiency and its compatibility with standard web browsers.
- Policy Compliance: Any modifications to institutional lending policies will necessitate corresponding updates to the system's functional specifications.
- Time Tracking: Accurate system date/time is essential for correctly tracking overdue items and calculating penalties.
- Payment Gateway (Conditional): Integration with an external payment gateway will be required if the system is expanded to include online fine payment capabilities.

3. External Interface Requirements

3.1 User Interfaces

The Library Management System (LMS) provides distinct, role-specific web interfaces for its two primary user groups: Members (Students/Staff) and Librarians (Administrative Personnel). The system is architected to ensure a consistent, intuitive, and high-usability User Interface (UI) across all functionalities, minimizing the learning curve and facilitating efficient operation.

All user interfaces are delivered through standard web browsers and employ a responsive design methodology, guaranteeing full compatibility and optimal display on desktop, laptop, tablet, and mobile computing devices.

3.1.1 Member Interface

The Member Interface is designed to enable patrons to independently search, reserve, borrow, and manage their interactions with the library resources.

Key functional components include:

- Dashboard: A summary view displaying currently issued books, critical due dates, real-time overdue alerts, and pending financial obligations (fines).
- Resource Search & Discovery: A comprehensive module allowing users to search the collection by criteria such as title, author, ISBN, or subject category, along with the ability to browse curated collections.
- Book Details: Displays the availability status, total number of copies, author information, and the user's eligibility criteria for borrowing the specific resource.
- Issued Books Management: A detailed list of all resources currently on loan, including the issue date, definitive due date, and applicable fine calculation (if any).
- Fines & Payments: Presents overdue fines calculated by the system and facilitates the clearance of dues (contingent on the implementation of a payment gateway).
- Request Management: Allows users to place requests for resources currently unavailable or to submit suggestions for new acquisitions to the library catalog.
- Transaction History: A complete, archived log of all previously issued and returned resources associated with the member's account.
- User Profile: Enables the member to view account status and update personal contact and identifying details.

3.1.2 Librarian Interface

The Librarian Interface furnishes administrative staff with the necessary controls for the comprehensive management and oversight of the entire library system.

Key functional components include:

- Admin Dashboard: A high-level overview of system metrics, including total resource count, active member count, current outstanding issues, overdue items, and total accrued pending fines.
- Resource Management: Tools for the creation (Add), modification (Update), deletion (Delete), and categorization of library resources. This includes managing multi-copy inventory and resource availability status.
- Transaction Management: Core functionality for processing book issues to members and recording resource returns. This module automatically updates resource availability and initiates fine calculations.
- User Account Management: Functionality to view, verify, activate, and suspend member and staff accounts.
- Reporting & Analytics: Capability to generate critical operational reports, including:
    - Resource Inventory Status
    - Current Issued Resources
    - Overdue Resources List
    - Fine Collection Summary
- Fine Configuration Module: Allows for the configuration of fine parameters (e.g., daily rate, grace period) and the monitoring of overdue records and payment reconciliation.

3.1.3 Common Interface Features

The following design and functional features are uniformly applied across both Member and Librarian interfaces:

- Consistent Layout: Adherence to a standard visual structure comprising a persistent header, intuitive sidebar navigation, and a footer.
- Notifications and Alerts: Provision of clear, timely feedback through system messages (e.g., overdue warnings, success confirmations, error messages).
- Role-Based Access Control (RBAC): Strict enforcement of permissions ensuring users can only access functionalities appropriate to their defined role (Member vs. Librarian).
- Usability: Design emphasis on a simple, intuitive UI requiring minimal steps to execute core operational tasks.
- Responsiveness: Full optimization for adaptive display and functionality across various screen sizes and device types.

3.2 Hardware Interfaces

The LMS is designed to be hardware-agnostic and accessible via standard computing infrastructure.

- Client Device: Any computing device capable of running a modern, standards-compliant web browser (e.g., desktop PC, laptop, tablet, smartphone).
- Input Devices: Standard input mechanisms are supported: Keyboard, mouse, or touchscreen.

- Output Devices: Display screen for viewing; printer support (optional) for physical generation of reports or transaction receipts.
- Specialized Hardware: No proprietary or specialized hardware components are mandated for system operation.

3.3 Software Interfaces

The system is engineered to interoperate with a variety of standard software environments.

- Operating Systems (Client): Compatibility with mainstream operating systems including Windows, Linux, macOS, Android, and iOS.
- Web Browsers (Client): Full support for modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari.
- Database System (Server): Requires integration with a robust database management system, such as MySQL, PostgreSQL, or MongoDB.
- Backend Integration: Communication relies on RESTful APIs for the efficient management of core data entities: user accounts, resource catalog, transaction records, and fine data.

3.4 Communications Interfaces

The system employs a standard and secure client–server communication architecture.

- Architecture: Client–server model.
- Protocols: Communication is secured using HTTPS (Hypertext Transfer Protocol Secure) to ensure data confidentiality and integrity.
- Data Format: Data exchange between the client and server is standardized using the JSON (JavaScript Object Notation) format.
- Security: The use of HTTPS is mandatory for protecting sensitive user data and transaction information.

Additional Communication Capabilities:

- Real-time Updates: Supports near real-time status updates (e.g., instant availability changes upon issue/return, immediate fine accrual).
- Resilience: The communication layer is designed to gracefully handle network interruptions and incorporate retry mechanisms.
- Backend Interconnection: The server-side application communicates with the Database for persistent storage and, optionally, with a secure Payment Gateway for processing fine payments.
- Scalability: The architecture is designed to efficiently manage a high volume of concurrent user sessions and transactions.

4. System Features

The following section describes the major functional features provided by the Library Management System (LMS). Each feature includes a description, priority, stimulus/response sequence, and functional requirements.

4.1 Sign Up

4.1.1 Description and Priority

Allows a new user (student/staff) to register and create a library account.
Priority: High
Precondition: User is not registered

4.1.2 Stimulus/Response Sequences

1. (S) User selects Sign Up and enters details (name, ID, email, password).
   (R) System validates and creates accounts.

4.1.3 Functional Requirements

- REQ-4.1-1: System shall allow account creation with valid credentials.
- REQ-4.1-2: System shall validate input and show errors.

4.2 Login

4.2.1 Description and Priority

Allows registered users (Member/Librarian) to log in.
Priority: High
Precondition: User must be registered

4.2.2 Stimulus/Response Sequences

1. (S) User enters login credentials.
   (R) System verifies and grants/denies access.

4.2.3 Functional Requirements

• REQ-4.2-1: System shall authenticate users.
• REQ-4.2-2: Invalid login attempts shall be rejected.

4.3 Dashboard (Home)

4.3.1 Description and Priority

Displays overview of system activity.
Priority: High

4.3.2 Stimulus/Response Sequences

1. (S) User logs in.
   (R) Dashboard is displayed.

4.3.3 Functional Requirements

• REQ-4.3-1: Members see issued books and fines.
• REQ-4.3-2: Librarians see system stats.

4.4 View Book Categories

4.4.1 Description and Priority

Allows users to view categories of books.
Priority: High

4.4.2 Stimulus/Response Sequences

1. (S) User selects category section.
   (R) System displays categories.

4.4.3 Functional Requirements

• REQ-4.4-1: System shall display all categories.

4.5 View Book List

4.5.1 Description and Priority

Displays books under selected categories.
Priority: High

4.5.2 Stimulus/Response Sequences

1. (S) User selects a category.
   (R) The system shows a books list.

4.5.3 Functional Requirements

• REQ-4.5-1: System shall display book details and availability.

4.6 Search Book

4.6.1 Description and Priority

Allows searching books.
Priority: High

4.6.2 Stimulus/Response Sequences

1. (S) User enters keyword.
   (R) System shows results.

4.6.3 Functional Requirements

• REQ-4.6-1: Support search by title, author, category.

4.7 Issue Book

4.7.1 Description and Priority

Allows librarians to issue books.
Priority: High
Precondition: Book available

4.7.2 Stimulus/Response Sequences

1. (S) Librarian selects issue option.
   (R) System assigns book and due date.

4.7.3 Functional Requirements

• REQ-4.7-1: System shall assign due date.
• REQ-4.7-2: Update availability.

4.8 Return Book

4.8.1 Description and Priority

Allows librarians to accept returned books.
Priority: High

4.8.2 Stimulus/Response Sequences

1. (S) Book is returned.
   (R) System updates status and calculates fine.

4.8.3 Functional Requirements

• REQ-4.8-1: System shall update book status.
• REQ-4.8-2: System shall trigger fine calculation.

4.9 Book Management

4.9.1 Description and Priority

Allows librarians to manage books.
Priority: High

4.9.2 Stimulus/Response Sequences

1. (S) Librarian adds/edits book.
   (R) System updates database.

4.9.3 Functional Requirements

• REQ-4.9-1: Add, update, delete books.

4.10 View Issued Books

4.10.1 Description and Priority

Shows books issued to members.
Priority: High

4.10.2 Stimulus/Response Sequences

1. (S) User opens the issued section.

(R) The system displays books.

4.10.3 Functional Requirements

• REQ-4.10-1: Show due dates and status.

4.11 Book Request

4.11.1 Description and Priority

Allows users to request books.
Priority: Medium

4.11.2 Stimulus/Response Sequences

1. (S) User submits request.
   (R) System stores request.

4.11.3 Functional Requirements

• REQ-4.11-1: Store request details.

4.12 History Tracking

4.12.1 Description and Priority

Tracks past issued books.
Priority: Medium

4.12.2 Stimulus/Response Sequences

1. (S) User views history.
   (R) System displays records.

4.12.3 Functional Requirements

• REQ-4.12-1: Maintain history log.

4.13 Fine Calculation

4.13.1 Description and Priority

Calculates overdue fine.

Priority: High

4.13.2 Stimulus/Response Sequences

1. (S) Book is overdue.
   (R) The system calculates fine.

4.13.3 Functional Requirements

• REQ-4.13-1: Fine = overdue days × rate.
• REQ-4.13-2: Fine must update automatically.

4.14 Fine Payment

4.14.1 Description and Priority

Allows payment of fines.
Priority: Medium

4.14.2 Stimulus/Response Sequences

1. (S) User pays fine.
   (R) System updates status.

4.14.3 Functional Requirements

• REQ-4.14-1: Update payment record.

4.15 Reports Generation

4.15.1 Description and Priority

Generates reports.
Priority: High

4.15.2 Stimulus/Response Sequences

1. (S) Librarian selects report.
   (R) System generates reports.

4.15.3 Functional Requirements

• REQ-4.15-1: Generate inventory and issue reports.

4.16 User Management

4.16.1 Description and Priority

Manage users.
Priority: High

4.16.2 Stimulus/Response Sequences

1. (S) Librarian accesses users.
(R) System displays accounts.

4.16.3 Functional Requirements

• REQ-4.16-1: View and manage users.

4.17 Notifications

4.17.1 Description and Priority

Notify users about updates.
Priority: Medium

4.17.2 Stimulus/Response Sequences

1. (S) Event occurs.
(R) Notification shown.

4.17.3 Functional Requirements

• REQ-4.17-1: Show real-time alerts.

4.18 Logout

4.18.1 Description and Priority

Logs users out.
Priority: High

4.18.2 Stimulus/Response Sequences

1. (S) User clicks logout.
(R) Session ends.

4.18.3 Functional Requirements

- REQ-4.18-1: Invalidate session.

4.19 Book Availability Management

4.19.1 Description and Priority

Ensures that books can only be issued if copies are available and displays real-time availability status.
Priority: High

4.19.2 Stimulus/Response Sequences

1. (S) User views book details.
   (R) System displays availability status (Available / Not Available).
2. (S) Librarian attempts to issue a book.
   (R) System checks availability before issuing.

4.19.3 Functional Requirements

- REQ-4.19-1: System shall track number of available copies.
- REQ-4.19-2: System shall prevent issuing when no copies are available.

4.20 Issue Limit per User

4.20.1 Description and Priority

Restricts the number of books a user can issue at a time.
Priority: High

4.20.2 Stimulus/Response Sequences

1. (S) Librarian tries to issue a book to a user.
   (R) System checks user's current issued count.
2. (S) Limit exceeded.
   (R) System denies issuing.

4.20.3 Functional Requirements

- REQ-4.20-1: System shall define a maximum issue limit per user.
- REQ-4.20-2: System shall block issuing beyond the limit.

4.21 Fine Restriction Control

4.21.1 Description and Priority

Prevents users with unpaid fines above a threshold from issuing new books.
Priority: High

4.21.2 Stimulus/Response Sequences

1. (S) User has a pending fine.
   (R) The system checks the fine amount.
2. (S) User tries to issue a book.
   (R) System blocks transaction if fine exceeds limit.

4.21.3 Functional Requirements

• REQ-4.21-1: System shall define fine threshold.
• REQ-4.21-2: System shall restrict issuing if threshold exceeded.

4.22 Book Renewal (Extension)

4.22.1 Description and Priority

Allows users to request extension of the due date for issued books.
Priority: Medium

4.22.2 Stimulus/Response Sequences

1. (S) User requests renewal.
   (R) System sends request to librarian.
2. (S) Librarian approves/rejects request.
   (R) System updates due date.

4.22.3 Functional Requirements

• REQ-4.22-1: System shall allow renewal requests.
• REQ-4.22-2: System shall update due date upon approval.

4.23 Book Copy Management

4.23.1 Description and Priority

Manages multiple copies of the same book.
Priority: High

4.23.2 Stimulus/Response Sequences

1. (S) Librarian adds book copies.

(R) System updates inventory count.

2. (S) Book is issued/returned.
   (R) System updates available copies.

4.23.3 Functional Requirements

- REQ-4.23-1: System shall maintain count of book copies.
- REQ-4.23-2: System shall update count dynamically.

4.24 Role-Based Access Control

4.24.1 Description and Priority

Ensures only authorized users can access specific features.
Priority: High

4.24.2 Stimulus/Response Sequences

1. (S) User logs in.
   (R) System assigns role (Member/Librarian).
2. (S) User accesses a feature.
   (R) System checks permissions.

4.24.3 Functional Requirements

- REQ-4.24-1: System shall restrict admin features to librarian only.
- REQ-4.24-2: System shall prevent unauthorized access.

4.25 Audit and Activity Logs

4.25.1 Description and Priority

Tracks system activities for monitoring and accountability.
Priority: Medium

4.25.2 Stimulus/Response Sequences

1. (S) User performs action (issue/return/edit).
   (R) System logs activity.
2. (S) Librarian views logs.
   (R) System displays activity history.

4.25.3 Functional Requirements

- REQ-4.25-1: System shall log all critical actions.

• REQ-4.25-2: Logs shall be accessible to admin only.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The Library Management System shall be responsive and efficient during normal usage. The system must support multiple users (members and librarians) simultaneously without noticeable performance degradation. Core operations such as searching books, issuing and returning books, updating records, and calculating fines shall be processed quickly to ensure smooth interaction. The system shall update book availability, transaction status, and fine calculations in real time or near real time.

5.2 Safety Requirements

The system shall ensure the safety and integrity of stored data. All information related to users, books, transactions, and fines shall be protected against data loss, corruption, and unauthorized modification. Proper validation mechanisms shall be implemented to prevent incorrect data entry (e.g., invalid issue/return operations). Backup mechanisms shall be in place to recover data in case of system failure.

5.3 Security Requirements

The Library Management System shall implement secure authentication and authorization mechanisms to ensure that only authorized users can access the system. Sensitive data such as user credentials shall be securely stored (e.g., encrypted passwords). Role-based access control shall restrict functionalities (e.g., only librarians can issue or modify books). The system shall protect against unauthorized access, data breaches, and misuse of system resources.

5.4 Software Quality Attributes

- Usability: The system shall provide a simple and user-friendly interface that can be used easily by

students, staff, and librarians without technical expertise.
- Reliability: The system shall operate consistently without failures and ensure accurate processing of transactions such as issue, return, and fine calculation.
- Availability: The system shall be accessible 24/7 with minimal downtime.
- Scalability: The system shall support increasing numbers of users, books, and transactions without performance degradation.
- Maintainability: The system shall be designed in a modular way to allow easy updates, bug fixes, and feature enhancements.
- Efficiency: The system shall optimize resource usage and provide fast response times for all operations.

5.5 Business Rules

- The system shall enforce role-based access control, where members and librarians have access only to their respective functionalities.
- A member shall not be allowed to issue more books than the predefined issue limit.
- A book shall not be issued if no copies are available.
- Late fines shall be calculated automatically based on overdue duration, and users with unpaid fines beyond a threshold may be restricted from issuing new books.
- Book renewal requests shall require approval from the librarian before extending the due date.
- All transactions (issue, return, fine payment) shall be recorded and maintained for auditing purposes.
- Only librarians shall have permission to add, update, or delete book records and manage user accounts.

6. Other Requirements

6.1 Database Requirements

The Library Management System shall use a centralized database to store user accounts, book details, categories, issue/return transactions, fine records, and audit logs. The database shall support efficient searching, filtering, and retrieval of records while maintaining data consistency and integrity. It shall prevent duplicate entries (e.g., duplicate book records or user accounts) and ensure accurate tracking of book availability and transaction history.

6.2 Internationalization Requirements

The system shall be designed to support future multilingual and localization features. All text elements such as labels, messages, and notifications shall be configurable so that additional languages can be integrated without major system redesign. Initially, the system may operate in a single language.

6.3 Legal and Ethical Requirements

The system shall comply with applicable data protection and privacy regulations. User information shall be collected and used only for library management purposes. Access to user data shall be restricted based on roles, and sensitive data shall be protected from misuse. All system activities, including issue/return transactions and fine handling, shall follow ethical and transparent practices.

6.4 Data Retention and Deletion Policy

The system shall maintain records of issued books, returns, fines, and user activities for a defined period. Old or inactive records may be archived or deleted based on institutional policies. User accounts and personal data shall not be stored indefinitely and shall be removed or anonymized when no longer required. Audit logs shall be retained for monitoring and accountability purposes.

6.5 Reuse Objectives

The system shall be designed to be reusable across different institutions such as colleges, schools, and organizations. Core components including user management, book management, transaction handling, fine calculation, and reporting shall be modular and reusable with minimal modifications.

6.6 Dependency Requirements

The operation of the Library Management System depends on a stable internet connection and access to modern web browsers. The system relies on database services for data storage and retrieval. Certain functionalities such as fine payment (if online) may depend on external payment gateways. System performance may also depend on server capacity and network conditions.

—-----END OF SOFTWARE REQUIREMENT SPECIFICATIONS—-----